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SIGNALS & SIGNALS

LABORATORY EXERCISE #2 – ANALOGUES
SIGNS

2018-2019 CEC

LABORATORY EXERCISE 2

ANALOG SIGNALS

Continuous-time signals or **analog signals** are signals whose independent variable varies over a continuous range of values. In one-dimensional signals, the domain of the signal is the interval of real numbers.

More simply, we can say that in continuous time signals the independent variable t is continuous, i.e. these signals are defined for any value of the independent variable. The dependent variable, i.e. the amplitude of the signal, is also continuous. An example of such signals is speech as a function of time. An analog signal is described as a function $x(t)$, where t is a real number.

MATLAB and in general in computing we work with discrete signals. However, one can easily approximate analog signals or functions, using discrete-time equivalent functions with a very small step size. In the following example we plot the analog signal $y(t) = \cos(t)$

y t .

```
>> t = 0:0.01:10;
```

```
>> y = cos(t);
```

Using the operator `:` with three arguments produces the following vector for the independent variable of time:

```
t = {0.00, 0.01, 0.02, ..., 9.98, 9.99, 10.00}
```

From here on out we have referred to functions of this type as "continuous", although in reality they are discrete-time with the assumption that the magnitude of the independent variable is chosen sufficiently small.

The following sequence of commands results in the plotting of the function $\cos(t)$,

```
>> t = 0:0.01:10;
```

```
>> y = cos(t);
```

```
>> plot(t,y)
```

To plot multiple plots of different functions on the same plot, use the subplot command .

Try the following command sequence:

```
t = 0: .01:5;
y1 = 1 - exp(-t).*sin(10*t);
y2 = exp(-t).*sin(10*t);
y3 = 1 - exp(-t).*sin(5*t);
y4 = exp(-t).*sin(5*t);
subplot(221)
plot(t,y1), xlabel('Time (sec)')
subplot(222)
plot(t,y2), xlabel('Time (sec)')
subplot(223)
plot(t,y3), xlabel('Time (sec)')
subplot(224)
plot(t,y4), xlabel('Time (sec)')
subplot(111)
title('Decaying Sinusoids')
```

Try to plot all 4 graphs in one line only using the **plot** command properly.

YOUR BASIC SIGNS

In this section, we present the basic symbols used in the design of systems and algorithms and how to construct and design them in Matlab.

1. Sinusoidal signals

The first basic category is the sinusoidal signals. These signals are of the form $(A) \cos(\omega t + \phi)$, where A is called the amplitude, ω is called the angular frequency, ϕ is called the phase. As is known, signals of this form are periodic and the period T is called the periodic period and is given by the form $T = 2\pi / \omega$. A useful quantity is ω

$$\omega /$$

frequency f of the sinusoidal wave. For frequency f , f is

$$\frac{1}{T}$$

$$\omega = 2\pi f$$

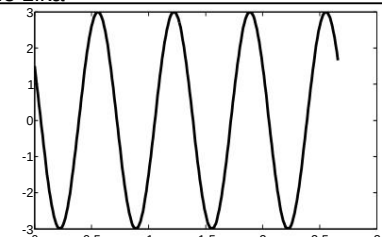
Example: To plot the equation $x(t) = 3\cos(3t)$ in 4 periods

$$x(t) = 3\cos(3t)$$

First we calculate the period as

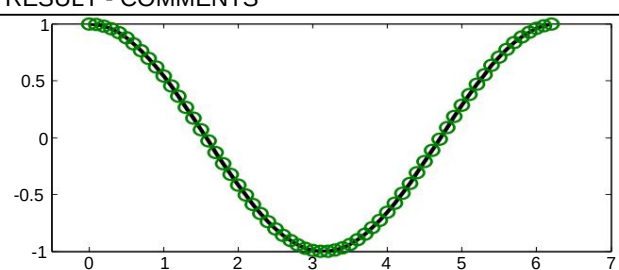
$$T = \frac{2\pi}{\omega} = \frac{2\pi}{3}$$

. In Matlab we have:

INTELLECTUAL	RESULT- COMMENTS
PROPERTY >> A=3; The number of digits is 3.	
>> omega=3*pi; The first term is 3 >> theta=pi/3; The second term is >> ω .	
T=2/3; The third term is 2/3.	$\omega / 3$.
>> t=0:0.01:4*T; Reduce the value t from 0 to 4T.	
>> x=A*cos(omega*t+thita); Oxidize the zika	
>> plot(t,x)	

When we talk about sinusoidal signals, we include the sinusoid and the cosinoid. To make the above a little more understandable, let's work a little with the phase of a sinusoid signal. We will draw the $\cos(t)$ signals in the same diagram.

and $\sin(t + \pi/2)$.

DNTOLDS	RESULT - COMMENTS
<pre>>> t=0:0.1:2*pi; >> x1=cos(t); >> x2=sin(t+pi/2); >> plot(t,x1,t,x2,'o')</pre>	 If we say that $\cos(t)$ and $\sin(t + \pi/2)$ are equal, then we have the following idea. The $\cos(t)$ is the same as the $\sin(t + \pi/2)$ is the same as the t. The t is the same as the t. The t is the same as the t. The t is the same as the t. The t is the same as the t. The t is the same as the t. $\omega = \frac{2\pi}{T}$. Axa kpnxnyke la zpkpexethexokaziee zia dpn zikaa ke yn ide ixopn.

Come on, I'm sick, I'm sick, I'm sick

Ae^{it} The night will come.

It is a place where you can go from home. $T = \frac{2\pi}{\omega}$. From the beginning of the world

Euler's equation is:

$$Ae^{it} = \cos(t) + j\sin(t)$$

From the beginning of the Re{

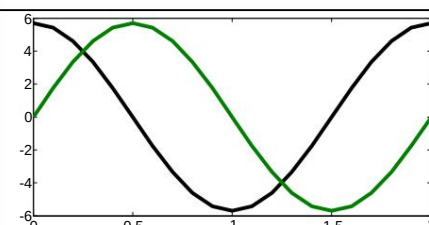
Ae^{it} } $\cos(t)$ Re{z} $\sin(t)$ Im{z}

The first one is the one that is missing z and the second is the one that is missing z.
one that is missing z.

Example: To calculate the cost of the well () 2 ft of income yt

$e^{it/3}$ The price is too high.

It is absolutely necessary to go outside. $T = \frac{2\pi}{\omega}$ 2. Which Matlab program should I use?

DNTOLDS	RESULT-COMMENTS
<pre>>> t=0:1:2; >> y_re=real(2*exp(j*pi*t+pi/3)) >> y_im=imag(2*exp(j*pi*t+pi/3));</pre>	Oxidized and the extracted and the extracted oil () 2 ft yt yinp $e^{it/3}$.
<pre>>> plot(t,y_re,t,y_im);</pre>	 The average of the given values is the sum of the required values and the sum of the required values is 2cos(the sum of the $y = t\pi/4$) yes values and the sum of the values) is 2sin($y = t\pi/4$) .

2. The unitary step function

A very useful symbol that we encounter is that of the unit step function $ut()$ which in English is called **the unit step function**. The unit step function is defined as follows:

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

Since this is a continuous value at $t = 0$ for practical reasons and is omitted here, we define the unitary step function as:

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

The function that gives us the step in Matlab is the **heaviside(t)** function.

According to Matlab programmers, the unitary step function is defined as

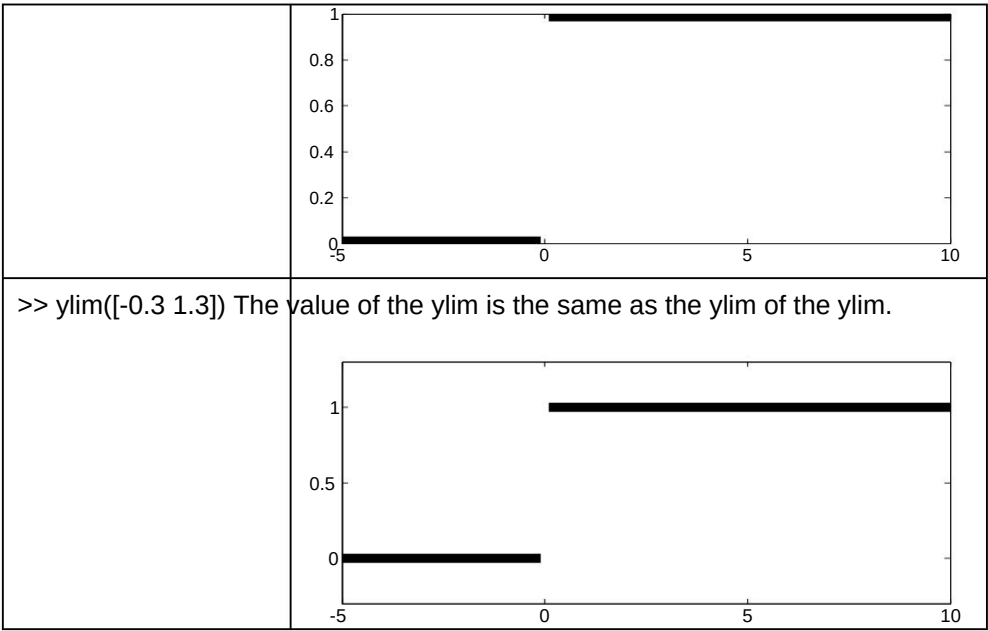
$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

The value is not defined at $t = 0$.

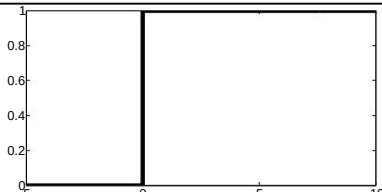
In the **example** below we give 3 ways to design the stepwise function.

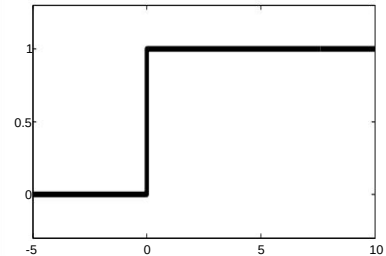
First method : Using the **Heaviside function**.

DNTOLDS	RESULT	COMMENTS
>> t=-5:0.1:10;	t= -5.....,-0.1, 0, 0.1,...,10	Oxidation is a very important factor in the development of t from
>> u=heaviside(t) u=	0.... ,0, Nan , 1,.....,1	-5 to 10 Oxidation is a complex process $u(t)$
>> plot(t,u)	The middle is a beautiful place.	

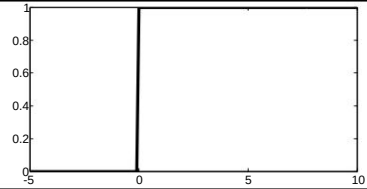
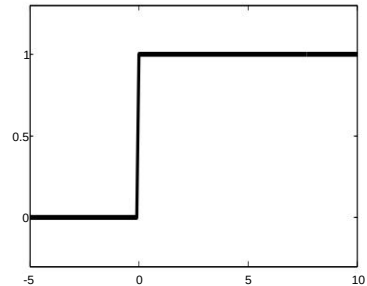


In a way: Design with the technical design of functions of many branches.

DNTOLDS	RESULT t1=	COMMENTS
>> t1=-5:1:0	-5 -4.90	Oxidation the of the first and second
>> t2=0:1:10	t2= 0 0.1.....10	molars the Oxidation of the second molar
>> u1=zeros(size(t1));	u1= 0 0.....0	The function ut() is called the Why is the t1
>> u2=ones(size(t2)); u2=	1 1.....1	The function ut() is called the The earth is red t2.
>> t=[t1 t2];	t= -5.....-0.1 0.....10	Packed lunch
>> u=[u1 u2];	u= 0.....0 1.....1	Paxazeze zplaxiezeol
>> plot(t,u)		He was getting sick.

<pre>>> ylim([-0.3 1.3])</pre>		The sun is shining brightly and the moon is shining brightly. yinp
--------------------------------------	--	---

Third way: Design a plan with a specific number of aces and zeros.

DNTOLDS	RESULT - COMMENTS	
>> t=-5:1:10	t=-5.....-0.1 0.....10	Oxidize the red blood cells.
>>u=[zeros(1,50)ones(1,101)];	u=0.....0 1.....1 The number of days in the t period is 151. The number of days in the t period is 50 and the number of days in the t period is 101 (the number of days in the t period is 0).	
>> plot(t,u)		He was a genius.
>> ylim([-0.3 1.3])		The most important thing is to be able to express your feelings and thoughts.

3. Impact function or Dirac function.

The Dirac structure $\delta(t)$ is defined mathematically through the two properties it has: The first properties are:

$$\int_{-\infty}^{\infty} \delta(t) f(t) dt = f(0)$$

If we define that $f(t) = y(t)$ the prefix transforms as:

$$y(t) = 1 \quad .$$

In practice, if we want to design the function $y(t)$, we resort to two definitions: The first one that even its programmers agree with Matlab is the equation:

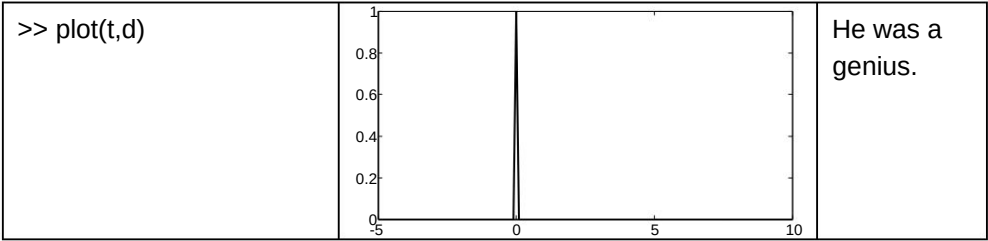
$$y(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases} .$$

The second definition is usually used when we have distinct signals and in this case it is preferable for the function $y(t)$ to be called the **Delta function**. For the Delta function the definition is:

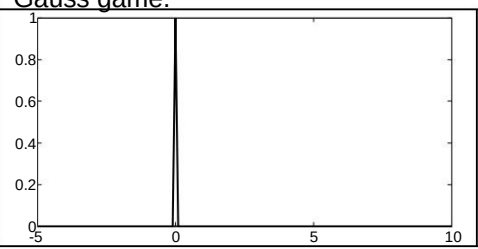
$$y(t) = \begin{cases} 1 & t = 0 \\ 0 & t \neq 0 \end{cases} .$$

We note that if we use this definition for the design of the function, the scheme we obtain is closer to what we encounter in the mathematical theory of systems and systems. Putting it in Matlab we have:

DNTOLDS	RESULT-COMMENTS
>> t1=-5:1:-0.1;	Oxidize the first part of the equation Oxidize
>> t2=0;	the second part of the equation to get the same value of the equation, with the equation $t \geq 0$ >> t3=0.1:1:10; Oxidize the
second part of the equation	
>> d1=zeros(size(t1)); Set the value of the square root $y(t)$ for the value t1	
>> d2=1;	The value of the function $y(t)$ is given by the value t2.
>> d3=zeros(size(t3)); Set the value of the square root $y(t)$ to the value of t3.	
>> t=[t1 t2 t3];	Packed lunch
>> d=[d1 d2 d3];	Paxazeze zplaxiezeol

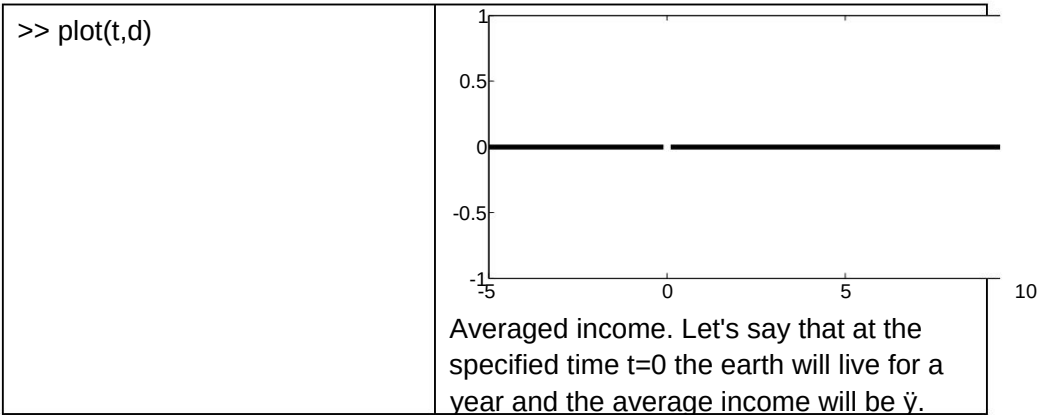


This form coincides with the definition of the **Gaussian pulse**. The function **gausspuls(t)** creates the same signal.

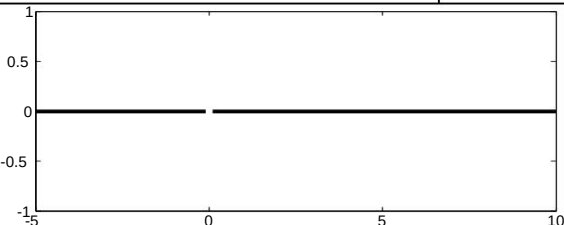
DTOLDS >>	RESULT-COMMENTS	
t=-5:.1:10; >>	Oxytocin Oxytocin	
s=gausspuls(t) >>	Gauss game.	
plot(t,s)		He was concentrating. yinp Zika.

In the case where we use the first definition (i.e. $y(t) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}$) Mrs. we have:

DNTOLDS	RESULT-COMMENTS	
>> t=-5:1:10	t=-5,...-0.1 , 0, 0.1 ,10	Oxidation e the very sad a
>> d = [zeros(1,50) inf zeros(1,100)];	d=0,.....0 ,inf, 0 ,.....0 The number of days in the t period is 151. The number of days in the t period is 50. The number of days in the t period is 10. The number of days in the t period is 10.	



In Matlab there is a ready-made function **dirac(t)**, which we can use to construct our signal.

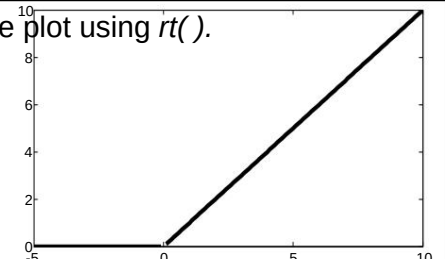
DNTOLDS	RESULT	COMMENTS
>> t=-5:1:10	t= -5,...-0.1 , 0, 0.1 ,10	Oxidation is a very common phenomenon. from -5 to 10
>> d=dirac(t);	d= 0,.....0 ,inf, 0 ,.....0	Oxidation of the surface ()t y
>> plot(t,d)	 He focused on the money. He explained that he had seen the poor man and the poor woman.	

4. The ramp (or climbing) function

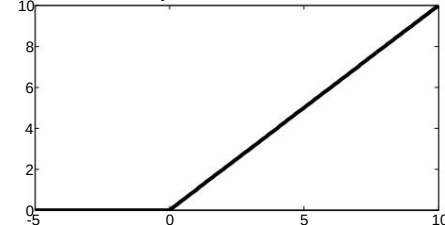
The ramp function $rt()$ is defined by the unit step width $ut()$ as follows:

$$rt() = \begin{cases} 0 & t < 0 \\ t & t \geq 0 \end{cases}$$

We continue to construct and use the unitary stepwise function as we did previously.

DNTOLDS APOTDLDSMA - COMMENTS	
>> t=-5:0.1:10; Oxidation of the heaviside t from -5 to 10 >> r=t.*heaviside(t);	
Oxidation of the heaviside $rt()$	tut
>> plot(t,r) Averages the plot using $rt()$.	

Another way to design the ramp function is with the multi-branch function design technique. So here we go:

DNTOLDS APOTDLDSMA- COMMENTS	
>> t1=-5:.1:-0.1; Oxidize the first row of data >> t2=0:.1:10;	
Oxidize the second row of data	
>> r1=zeros(size(t1));	The value of the square root is $rt()$ for the value t1
>> r2 = t2;	The function $rt()$ is defined for the time t2.
>> t=[t1 t2];	Packed lunch
>> r=[r1 r2];	Paxazeze zplaxiezeol
>> plot(t,r)	 The average of the square root of $rt()$.

5. The square wave

The square pulse $pT t()$ is a square pulse of width γ and time duration T. It is defined by the unit step width $ut()$ as follows:

$$pT t(\gamma) \gamma \gamma \gamma \left(\frac{T}{2} \right) \left(\frac{T}{2} \right)$$

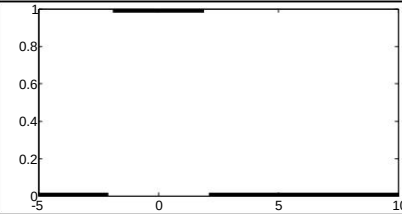
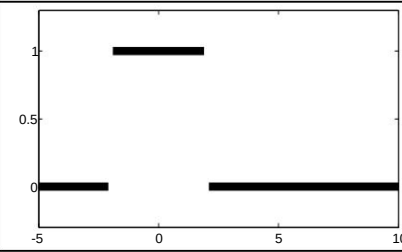
For the construction, we use two unitary step functions.

So, let's say we want to create a square pulse of duration $T = 4$

We want to construct the symbol $p_{T/4}(t)$. Substituting $T = 4$ in the definition we have:

$$p_{T/4}(t) = \begin{cases} 1 & -2 \leq t \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

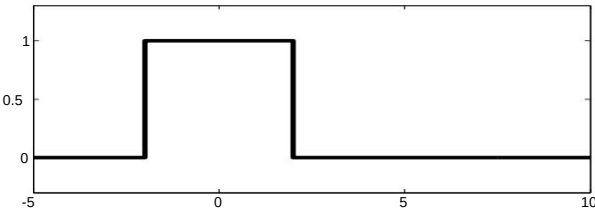
So in Matlab we say:

IMF	RESULT - COMMENTS	
>> t=-5:1:10;	t= -5,...,-2, -1.9,...2, 2.1,...10	Oxidation is a very common phenomenon. from -5 gain 10
>> u1=heaviside(t+2);	u1= 0,...,NaN, 1,...1	Oxidation of the $u(t+2)$
>> u2=heaviside(t-2);	u1= 0,...,0, 0,... NaN, 1...1	Oxidation of the $u(t-2)$
>> p=u1-u2;	p= 0,...NaN,1,...NaN, 0....0	The old game was played with a knife and a knife.
>> plot(t,p)		Wednesday the was a great day for the game.
>> ylim([-0.3 1.3])		The sun is shining brightly, and the sun is shining brightly.

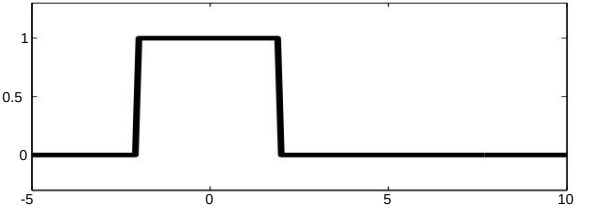
We assume that the signal $p_{T/4}(t)$ is a square pulse of width 1 and duration 4. The pulse is centered at $t = 0$.

Another way to design square wave signals is with the multi-branch function design technique. So here we have:

IMF	RESULT-COMMENTS
-----	-----------------

>> t1=-5:1:-2;	Oxidize the surface of the red clay t1.
>> t2=-2:1:2;	Oxidize the second most common type of t2.
>> t3=2:1:10;	Oxidizes the hexadecimal number t3.
>> p1=zeros(size(t1));	The normal distribution $pT(t)$ is given for the time t1. oh my
>> p2=ones(size(t2));	The normal distribution $pT(t)$ is given for the time t2. oh my
>> p3=zeros(size(t3));	The normal distribution $pT(t)$ is given for the time t3. oh my
>> t=[t1 t2 t3];	Packed lunch
>> p=[p1 p2 p3];	Paxazeze zplaxiezeol
>> plot(t,p); >> ylim([-0.3 1.3]);	 It centered the income and breathed the y-wave to express the income.

A third way to construct a square pulse signal is through the function `rectplus(t)` . The function `rectplus(t)` constructs a square pulse of duration $T=1$. If we call a different duration we give the command `rectplus(t,T)`, where T will be the corresponding duration. For the signal $pt\ 4()$ we therefore have:

DTOLDS >>	RESULT-COMMENTS
t=-5:1:10; >>	Oxidative stress
s=rectpulse(t,4);	The most important thing in life is to be happy.
>> plot(t,s) >> ylim([-0.3 1.3])	 It averages the income and releases the y-value to reflect the income.

VALIDATION OF SIGNALS

In this section we present the main properties that can characterize a symbol.

1. Periodic signals

A proportional signal $x(t)$ is called **periodic** if there is a constant number, T , for which $x(t) = x(t + T)$ is satisfied for every value of t . The constant number T is called period. An example of a periodic signal is the sinusoidal signal that has a period

$$x(t) = A \cos\left(\frac{2\pi}{T} t\right)$$

Example: Calculate whether the sine $x(t) = \sin(t/2)$ is periodic.

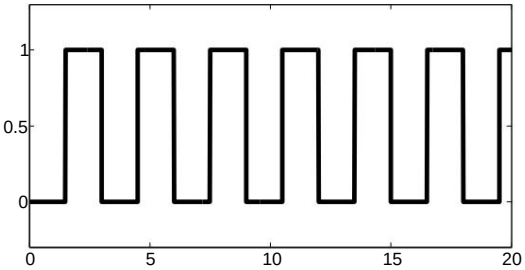
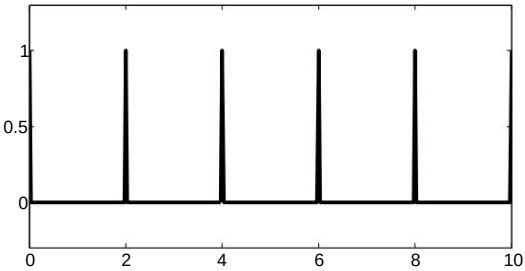
Initially we calculate the period as $T = \frac{2\pi}{\omega} = \frac{2\pi}{1/2} = 4\pi$. As is understandable, it is not

we can check the relation $x(t) = x(t + kT)$ for all integers k . We will examine whether it holds for $k=1$ and for time $t=1$.

COMMANDS	RESULT-COMMENTS	
t=1:5; x=sin(t)	x = 0.84 0.91 0.14 -0.76 -0.96 Define the signal at time 1 to 5	
T=2*pi;	Define the period	
>>for k=1:10 xk(k,:)=sin(t+k*T) end	With a for-loop we create a table where each row has the values of $x(t + kT)$ for $k=1$ to 10.	
>>xk	xk = 0.84 0.91 0.14 -0.76 -0.96 0.84 0.91 0.14 -0.76 -0.96 0.84 0.91 0.14 -0.76 -0.96 0.84 0.91 0.14 -0.76 -0.96 0.84 0.91 0.14 -0.76 -0.96 0.84 0.91 0.14 -0.76 -0.96 0.84 0.91 0.14 -0.76 -0.96 0.84 0.91 0.14 -0.76 -0.96 0.84 0.91 0.14 -0.76 -0.96 0.84 0.91 0.14 -0.76 -0.96	xk is the matrix (10×5) $x(t + T)$ $x(t + 2T)$ $x(t + 3T)$... $x(t + 10T)$ We assume that all lines are equal in their distance and also in the initial symbol $x(t)$. Therefore, the symbol is periodic.

- Construction and design of periodic signals.

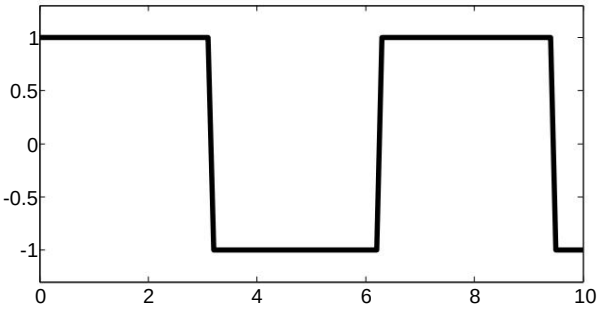
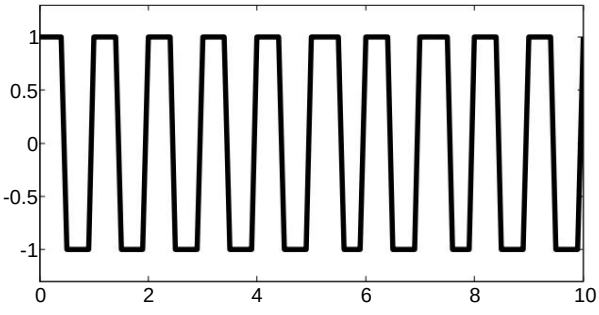
So far, the signals we have constructed (e.g., the square pulse) have been defined and designed for a time of one period. In Matlab, there are built-in functions that allow us to construct periodic functions. The first such function that we will consider is the **gensig** function. Its syntax is **[s, t]=gensig(type, T, t, t_sampling)**, where type is the type of the periodic signal. Available signal types are: **'sin'** for the sine, **'square'** for the square pulse, and **'pulse'** for the Dirac pulse. The **T** parameter corresponds to the period of the signal, i.e. how many times our signal is repeated. **t** is the total time that we define the signal. While **t_sampling** is the sampling time, a concept that will become more clear in later chapters. If omitted, it takes the default value of 0.001. The signal is output at output **s** and the time at output **t**.

COMMANDS	RESULT- COMMENTS
>> [s,t]=gensig('square',3,20,0.01)	Square pulses repeated with period T=3 for time t=20 and sampling time 0.01.
>> plot(t,s) >> ylim([-0.3 1.3])	 Drawing the appropriate axes. We see that at time 0 20 $\ddot{y} \ddot{t}$ we have square pulses that repeat with period T=3.
>> [s,t]=gensig('pulse',2,10);	Pulses repeating with period T=2 for time 0 10 $\ddot{y} \ddot{t}$ with
>> plot(t,s) >> ylim([-0.3 1.3])	 Design of suitable axes.

Another useful function is the **square** function. The square function essentially produces a squared sine that takes values of $\ddot{y} 1$. It is written in the same way as the sin function. For example, the command **square(t)** produces a squared sine

period $T \ddot{y} \frac{2\ddot{y}}{\ddot{y}} 2\ddot{y}$.

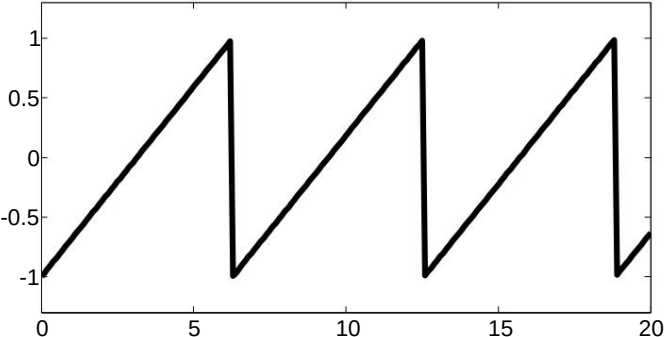
COMMANDS	RESULT-COMMENTS
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<pre>>> t=0:0.1:10;</pre>	We define time
<pre>>> s=square(t);</pre>	Definition of the word.
<pre>>> plot(t,s); >> ylim([-1.3 1.3])</pre>	 We assume that our signal has period $T \approx 2$ and its width takes values ≈ 1.
<pre>>> s2=square(2*pi*t);</pre>	Definition of the word "simatos" at the same time.
<pre>>> plot(t,s2); >> ylim([-1.3 1.3]) ;</pre>	 We are encouraged that our brand is still alive. period $T \approx \frac{2\pi}{\omega} \approx \frac{2\pi}{2\pi} = 1$

A similar function is the **sawtooth function**. The **sawtooth** function produces a "sawtooth" sine wave. It is written like the sin function. Command line

sawtooth (t) produces a "sawtooth" period sound

$$T \approx \frac{2\pi}{\omega} \approx \frac{2\pi}{2\pi} = 1$$

COMMANDS	RESULT-COMMENTS
<pre>>> t=0:0.1:20; >> s=sawtooth(t);</pre>	Definition of triangular pulses in the interval $0 \leq t \leq 20$.
<pre>>> plot(t,s); >> ylim([-1.3 1.3])</pre>	 We assume that the period is $T \approx 2$ and the amplitude ≈ 1.

Up until now, the signals that we have repeated periodically in time have had a very specific form (e.g., square). Let us suppose that we want to repeat in time for a specific number of periods the signal

$$x(t) = \begin{cases} 1 & 0 \leq t < 1 \\ 0 & 1 \leq t < 2 \end{cases}$$

The solution to this problem is provided by the **repmat** function. Its syntax is **xp=repmat(x,N,K)**. This command means that we take the vector **x** and iterate it over **N** rows and **K** columns. The result is stored in **NxK**

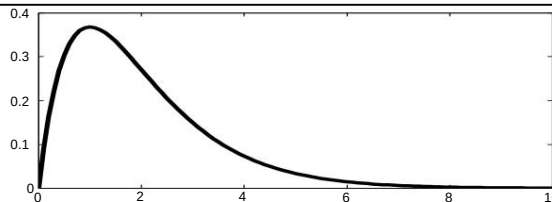
xp board .

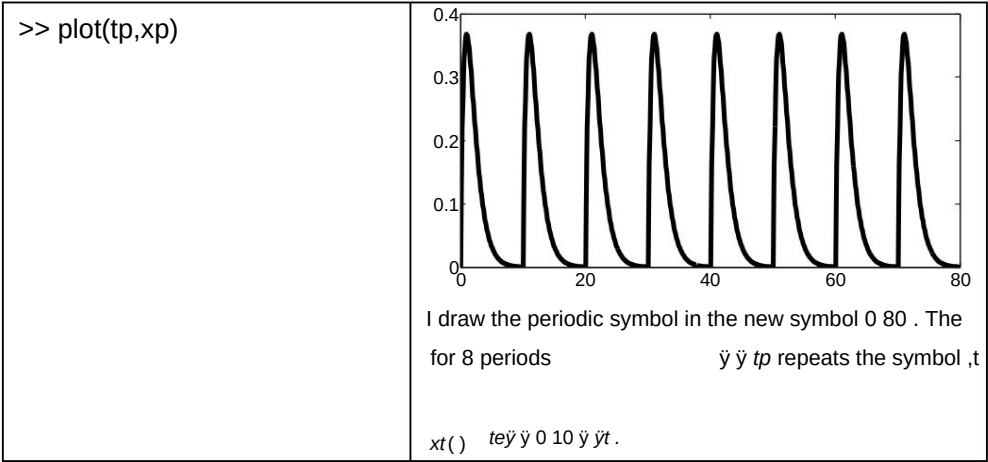
The **repmat** command is presented in the table below:

COMMANDS	RESULT-COMMENTS	
>>x=[1 2]; >>N=3; K=4;	We define the basic symbol x and the number of rows and columns.	
>> xp=repmat(x,N,K)	<pre> xp = 1 2 1 2 1 2 1 2 1 2 1 2 1 2 1 2 1 2 1 2 1 2 1 2 </pre>	We see that the matrix xp it is a $\begin{bmatrix} 1 & 2 \end{bmatrix}$ repeat of distribution in N x lines and K styles.

Returning to our original problem, let's iterate over time for life $\ddot{y}$ we work until 0 10

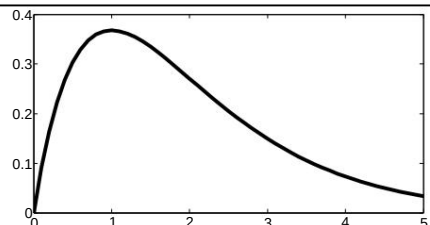
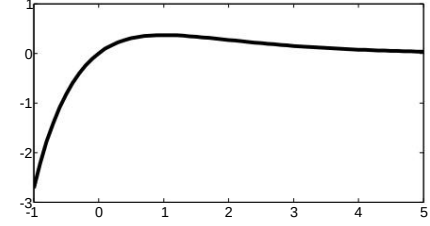
specific number of periods (say 8) the symbol () equals: $x(t) = \begin{cases} 1 & 0 \leq t < 1 \\ 0 & 1 \leq t < 2 \end{cases}$, t

COMMANDS	RESULT-COMMENTS
>> t=0:0.1:10; >> x=t.*exp(-t); >> plot(t,x);	 We define the value x in time of one period (here T=10)
>> xp=repmat(x,1,8);	I repeat the line x in 1 line and 8 lines. I find the new time when I
>> tp=linspace(0.80,length(xp))	draw xp, dividing the time into pieces equal to the number 0 $\ddot{y}$ tp 80 of the elements of xp.



2. Causal signals

A signal is called **causal** if it is zero for negative values of time , i.e.: . In the opposite case, the signal is called uncausal.

COMMANDS	RESULT	COMMENTS
<pre>>> t1=0:0.1:5; >> x1=t1.*exp(-t1); >> plot(t1,x1)</pre>		Causal sima
<pre>>> t2=-1:0.1:5; >> x2=t2.*exp(-t2); >> plot(t2,x2)</pre>		No causal link

3. Even and Odd Signals

A number is said to be **even** (presents even symmetry) when: Conversely, it is said to be odd (presents odd symmetry) when: .

Example: We will examine the signals $xt() \ddot{y} \ddot{y} t^2$ and $y t() \ddot{y} t^3$ Because it is not possible examine the two signals in time (,) $\ddot{y} \ddot{y}$ initially and define two time intervals in which we examine our two signals . Then we check the values of the error for these intervals.

COMMANDS	RESULT	COMMENTS
>> t1=0:4	t1 = 0 1.00 2.00 3.00 4.00	The time t
>> t2=0:-1:-4	t2 = 0 -1.00 -2.00 -3.00 -4.00	The time -t
>> x1=t1.^2	x1 = 0 1.00 4.00 9.00 16.00	The xt()
>> x2=t2.^2	x2 = 0 1.00 4.00 9.00 16.00	that To xt It is assumed xt() is so the xt() is even
>> y1=t1.^3	y1 = 0 1.00 8.00 27.00 64.00	The yt() .
>> y2=t2.^3	y2 = 0 -1.00 -8.00 -27.00 - 64.00	yt() is . enough that equal() so() ytyt is odd. t is y t()

ÿ, All signals can be expressed as the terminus of an even number, xt e ÿ and one unnecessary strings, , as , where: and

4. Energy Signals - Earth Signals

One of the most important signal distinctions is that between **energy signals** and **ground signals** .

For a given signal, , the energy is given by the relation:

$$E_x = \lim_{T \rightarrow \infty} \int_{-T}^T |x(t)|^2 dt = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

where is the measure of the string.

A signal is characterized as an energy signal when its energy is finite in the range $0 \leq E_x < \infty$.

The length of the string is given by the relation:

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T |x(t)|^2 dt$$

If the signal is periodic, then the frequency is given by the relation:

$$P_x = \frac{1}{T} \int_{t_0}^{t_0+T} |x(t)|^2 dt$$

A line is characterized as a straight line when $0 < P_x < \infty$.

At this point, in order to calculate the force and energy of a wire in Matlab, we must use the **limit command-function**. The **limit** function is used only with symbolic expressions. Its syntax is **limit(F,x,a)**

,where it calculates the limit of the symbolic expression **F** when the symbolic variable **x** tends to **a**.

An example of a linear action is $x(t) = t$ (expression) an example of a string () 1)
 $x(t) = t$ is

Let's start from the equation $x(t) = t$.

COMMANDS	RESULT-COMMENTS
>> syms t T	Definition of symbolic variables
>> x=heaviside(t);	First we define the symbol $x(t)$.
>> d=int(abs(x)^2,t,-T,T);	We first calculate the $\int_{-T}^T x(t) ^2 dt$
>> Ex=limit(d,T,inf)	Ex = Inf Calculation this energy $E_x = \lim_{T \rightarrow \infty} \int_{-T}^T x(t) ^2 dt$ We see that $\lim_{T \rightarrow \infty} x(t) ^2$ $E_x \rightarrow \infty$ so $x(t)$ is not equal energy.
>> Px=limit((1/(2*T))*d,T,inf)	Px = 1/2 Calculation this is

	$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \dot{y}^2 dt$ We see that $P_x = 0.5 P$ so $u(t)$ is equal to Ishphos.
--	--

For the equation $\ddot{y} = u(t)$ we have:

COMMANDS	RESULT-COMMENTS
<code>syms t T</code>	Definition of symbolic variables >>
<code>xtut(t) = heaviside(t) - heaviside(t-1);</code>	First we define the symbol $x(t)$.
<code>>> d = int(abs(x)^2, t, -T, T);</code>	For convenience, we first calculate $\int_{-T}^T \dot{y}^2 dt$.
<code>>> Px = limit((1/(2*T))*d, T, inf)</code>	$P_x = 0$ Calculation see We $P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \dot{y}^2 dt$ that $P_x = 0$ so $u(t)$ is not equal Ishphos.
<code>>> Ex = limit(d, T, inf)</code>	$E_x = 1$ Calculation this energy $E_x = \int_{-T}^T \dot{y}^2 dt$ We see that lim $E_x = 1$ so $u(t)$ is $(\ddot{y})$ energy levels.

5. Causal signals - random signals

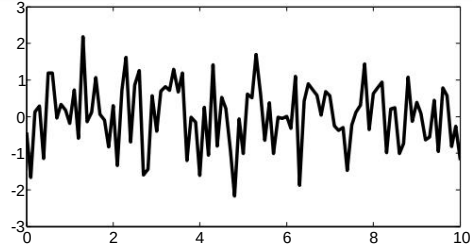
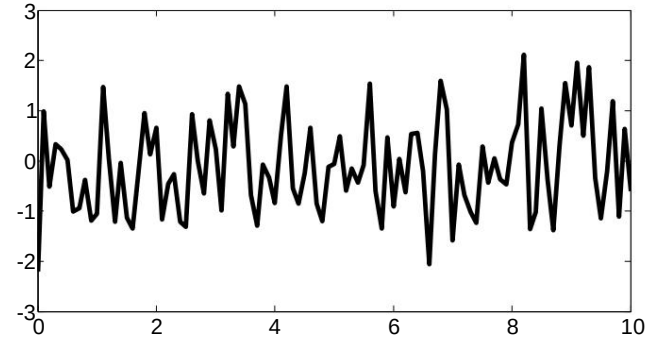
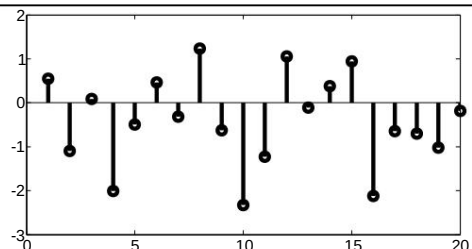
Another important distinction between signals is that between deterministic and random signals. When the values that a signal takes at any point in time are defined without uncertainty, the signal is characterized as a **deterministic signal** or a **random signal**.

In practice, we encounter many signals whose value at any given time cannot be predicted with certainty. These signals are called random or stochastic signals.

An example of deterministic signals is the unitary step function and the thesaurus signals for which for any time t we know what their values are.

An example of a random number is the kerma curve. The kerma curve is commonly modeled as a sequence of random numbers that follow the

normal distribution with mean value $\bar{y} = 0$ and variance $\bar{y}^2 = 1$. In the following example we draw such a line.

COMMANDS	RESULT-COMMENTS
<code>t=0:0.1:10; >></code>	We define a time interval >>
<code>x1=randn(size(t));</code> We define a set of random numbers with the width of the randn function.	
<code>>> plot(t,x1);</code>	 We assume that the signal is $x_1(t)$ is complete random and does not depend on the independent variable t.
<code>>> t=0:0.1:10;</code> <code>>> x2=randn(size(t));</code> <code>>> plot(t,x2);</code>	 Repeating the process for the same time we get a different equally random number $x(t)$. Remember that in no case can we predict the value of a random number.
<code>>> x=randn(20,1);</code> <code>>> stem(x);</code>	 Random discrete-time signal

VALIDATION OF SIGNALS

1. Signal to time conversion

Many times in practice, symbols are presented that are related to each other by changing the independent variable, namely time. In this section, we will see the **basic transformations – conversions** that we can make to a symbol by “experimenting” with time.

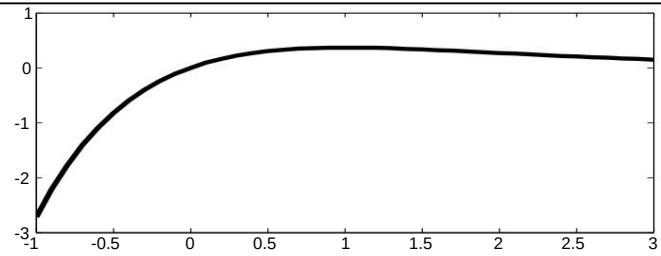
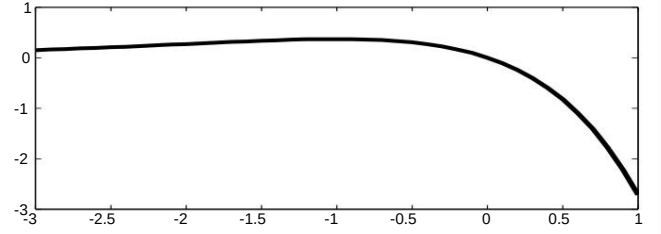
• Reflection

The first transformation of a string as of the time we are examining is the

reflection. We say that a line $y(t)$ is the reflection of the line $x(t)$ as if when:

The line $y(t)$ is essentially the symmetric line $x(t)$ about the x axis. The reflection transformation results in the alternation of the “negative” and “positive” time of a line. and

I live in Istanbul, I lived () in the $x(t)$, - $y(t)$. To avoid mistakes following places:

IMF	RESULT - COMMENTS
<pre>>> t=-1:1:3; >> x=t.*exp(-t); >> plot(t,x);</pre>	 Average the absolute value $x(t)$.
<pre>>> plot(-t,x);</pre>	 Please help me. $y(t)$ $y(t)$ To determine the error $x(t)$ we need to find the value of the function t and then find the value of the function $x(t)$

• Time Scale Change

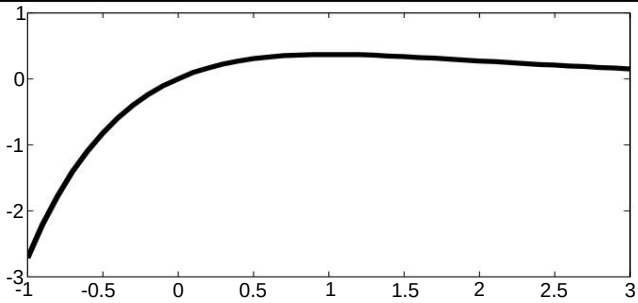
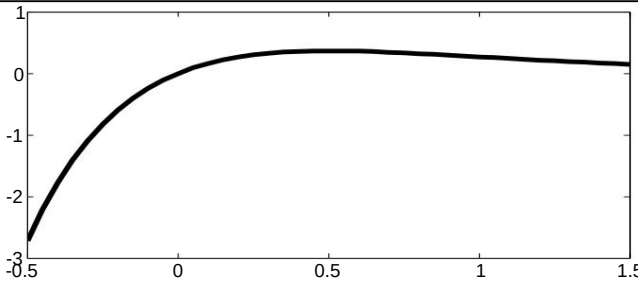
The second transformation of a signal with respect to time that we are considering is the **change in time scale**. We say that the signal constitutes a time systole of the signal when 1. This

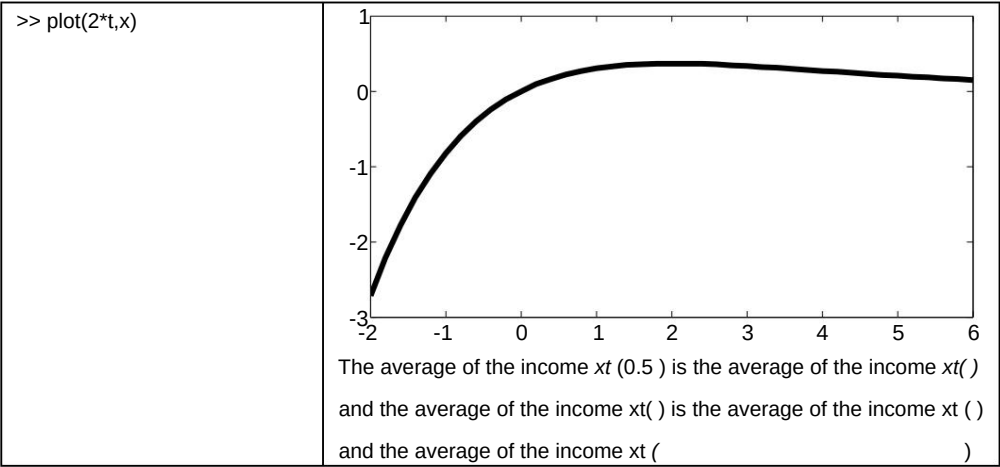
$x(at)$, practically means that the time duration of the signal decreases by a factor of $1/a$.

You are welcome to come and see the Zika virus in person 1. If you are interested in the Zika virus, please contact us.
the equation will be $1/a$.

I live in a city where I will live and play the game () for a long time: I will live the game (0.5)
(since $1/2$).

To plot in Matlab, I used the `ytz` function at the beginning of the plot () and called `plot(1/a)*t, x)`. I have seen that the plot is a function of the number of points in the t range, and that the t value is a function of the $1/a$ value .

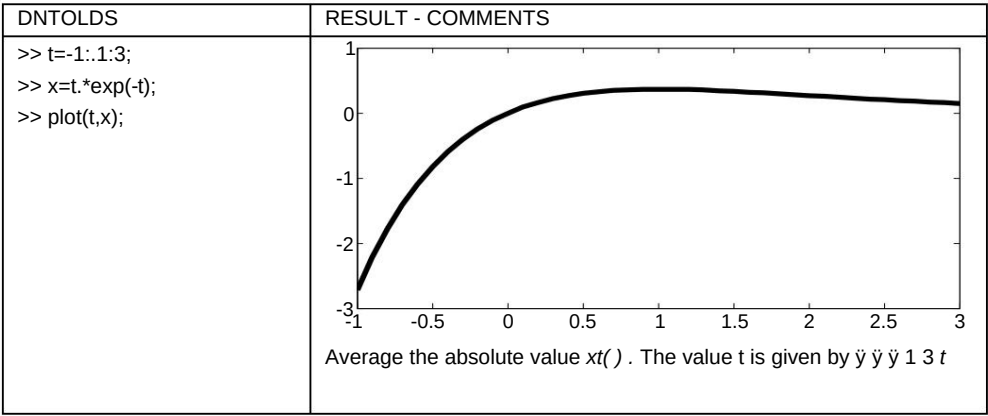
DNTOLDS	RESULT - COMMENTS
<pre>>> t=-1:1:3; >> x=t.*exp(-t); >> plot(t,x);</pre>	 Average the absolute value $xt()$. The value t is given by $1/3 t$
<pre>>> plot((1/2)*t,x)</pre>	 The average of the income $xt(2)$ is the average of the income $xt()$ and the average of the income $xt()$ is the average of the income $xt()$ and the average of the income $xt()$ income from the farm, the farm $0.5 1.5 t$

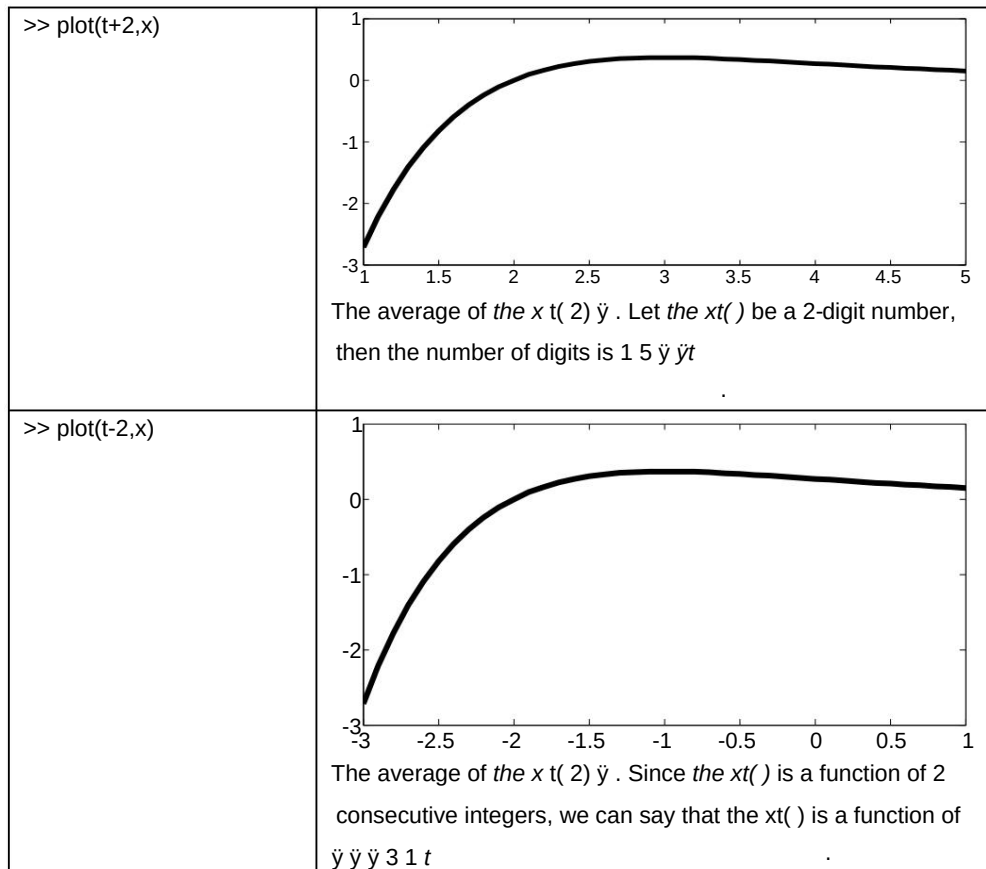


• Time Shift

The third transformation of a signal with respect to time that we consider is that of the **time displacement of the signal**. We say that a signal is a time displacement by signal $x(t)$ by $t \rightarrow 0$ the displacement $y(t)$ is a shift of the signal when $y(t)$ copy of the becomes t time units . When to the left (towards the right), and when $t \rightarrow 0$ the displacement towards $y(t)$,

As an example, let us assume that the time (t) is equal to the time $y(t)$, $1 \leq t \leq 3$. It will be $t \rightarrow y(t)$ (given that $t \rightarrow 2$) and the time (2) $(2) \rightarrow y(t)$ (given that $t \rightarrow 2$). Here we consider $x(t)$ to be a function called **plot(t+to,x)** .
If we use **plot(t-2,x)** to plot the $x(t)$ function and $2 x(t)$ functions (to plot the $x(t-2)$), from the beginning of the year. For the sake of the future, the year $y(t)$ then **plot(t+2,x)** to plot the $x(t)$ functions (to plot the $x(t)$ functions) and $2 x(t)$ functions (to plot the $x(t)$ functions) $t(2)$.





PRACTICE EXERCISES

ASK 1)

To solve the problem $x(t)$ the problem of the Heaviside equation.
b) the Heaviside equation.

ASK 2)

To destroy the zika $x(t)$

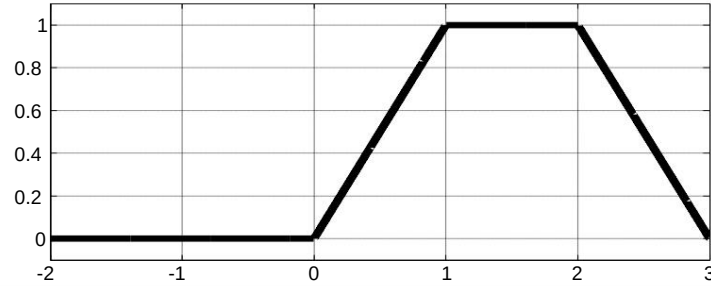
ASK 3)

To prepare the () exported game $t^3 \cos(10) 2(t)$, $pT(t)$, T.

ASK 4)

To be educated
yes la

It is a common
practice to use a
combination of
both.

**ASK 5)**

The value of the word: y you t , 0 y 5 t .

() is: • The value

of $x(t)$

- The first letter () x_t y y $x(t)$
- The Zika virus () x_t y y $x(t)$
- The azxnezka () $x_t + () x_t$.

ASK 6)

I live the life of a scoundrel y t y . To solve for () $\cos(2)$, 0 5 t y y

$x(t)$ scoundrel

- $x(t)$
- $x(t)$ y
- $x(t/5)$
- $x(1/3)$ t
- $x(1/3)$ t